Chapter 6, Problem 84CP

Problem

An 8-mH inductor is used in a fusion power experiment. If the current through the inductor is *i*(*t*) = 10 cos2 (πt) mA, for all *t* > 0 s, find the power being delivered to the inductor and the energy stored in it at *t* = 0.5 s.

Step-by-step solution

Step 1 of 3

The current through an 8 mH inductor is,

$$i(t) = 10\cos^2(\pi t)\ \text{mA}$$

The voltage across the inductor is,

$$v_L = L\frac{di_L(t)}{dt}$$

Substitute 8 mH for L, and $10\cos^2(\pi t)\ \text{mA}$ for i_L.

$$\begin{aligned} v_L &= (8\times10^{-3})\frac{d}{dt}\left[(10\times10^{-3})\cos^2(\pi t)\right] \\ &= (8\times10^{-3})(10\times10^{-3})\frac{d}{dt}\left[\cos^2(\pi t)\right] \\ &= (80\times10^{-6})\left[-2\pi\cos(\pi t)\sin(\pi t)\right] \\ &= -80\pi\sin 2\pi t\ \mu\text{V} \qquad (\text{since, } \sin 2x = 2\sin x\cos x) \end{aligned}$$

Step 2 of 3

Determine the power in the inductor.

$$p = v_L i_L$$

Substitute $10\cos^2(\pi t)\ \text{mA}$ for v_L and $-80\pi\sin 2\pi t\ \mu\text{V}$ for i_L.

$$\begin{aligned} p &= \left[(10\times10^{-3})\cos^2(\pi t)\right]\left[(-80\times10^{-6})\pi\sin(2\pi t)\right] \\ &= (10\times10^{-3})(-80\times10^{-6})\pi\cos^2(\pi t)\sin(2\pi t) \\ &= (-800\times10^{-9})\pi\cos^2(\pi t)\sin(2\pi t) \end{aligned}$$

Find the value of power delivered at $t = 0.5\ \text{s}$.

$$\begin{aligned} p(0.5) &= (-800\times10^{-9})\pi\cos^2\left[\pi(0.5)\right]\sin\left[2\pi(0.5)\right] \\ &= (-800\times10^{-9})\pi\cos^2\left(\frac{\pi}{2}\right)\sin(\pi) \\ &= 0 \end{aligned}$$

Thus, the power delivered to the inductor at $t = 0.5\ \text{s}$ is $\boxed{0}$.

Step 3 of 3

Determine the energy stored in the inductor.

$$
\begin{aligned}
w &= \int_0^t p\,dt \\
&= \int_0^{0.5} \left(-800\times10^{-9}\right)\pi\cos^2(\pi t)\sin(2\pi t)\,dt \\
&= \left(-800\times10^{-9}\right)\pi\int_0^{0.5} \frac{1+\cos(2\pi t)}{2}\sin(2\pi t)\,dt \\
&= \left(-800\times10^{-9}\right)\pi\int_0^{0.5} \left(\frac{\sin(2\pi t)}{2}+\frac{\sin(2\pi t)\cos(2\pi t)}{2}\right)dt
\end{aligned}
$$

Simplify the expression.

$$
\begin{aligned}
w &= \left(-800\times10^{-9}\right)\pi\int_0^{0.5} \left(\frac{\sin(2\pi t)}{2}+\frac{\sin(4\pi t)}{4}\right)dt \\
&= \left(-800\times10^{-9}\right)\pi\left[\frac{-\cos(2\pi t)}{2(2\pi)}+\frac{-\cos(4\pi t)}{4(4\pi)}\right]_0^{0.5} \\
&= \left(800\times10^{-9}\right)\pi\left[\frac{\cos(\pi)}{2(2\pi)}-\frac{\cos(0)}{2(2\pi)}+\frac{\cos(2\pi)}{4(4\pi)}-\frac{\cos(0)}{4(4\pi)}\right] \\
&= \left(800\times10^{-9}\right)\pi\left[-\frac{1}{4\pi}-\frac{1}{4\pi}+\frac{1}{16\pi}-\frac{1}{16\pi}\right]
\end{aligned}
$$

$$
\begin{aligned}
w &= -4\times10^{-7} \\
&= -0.4\ \mu\text{J}
\end{aligned}
$$

Thus, the energy stored in the inductor at $t = 0.5\ \text{s}$ is $\boxed{0.4\ \mu\text{J}}$.